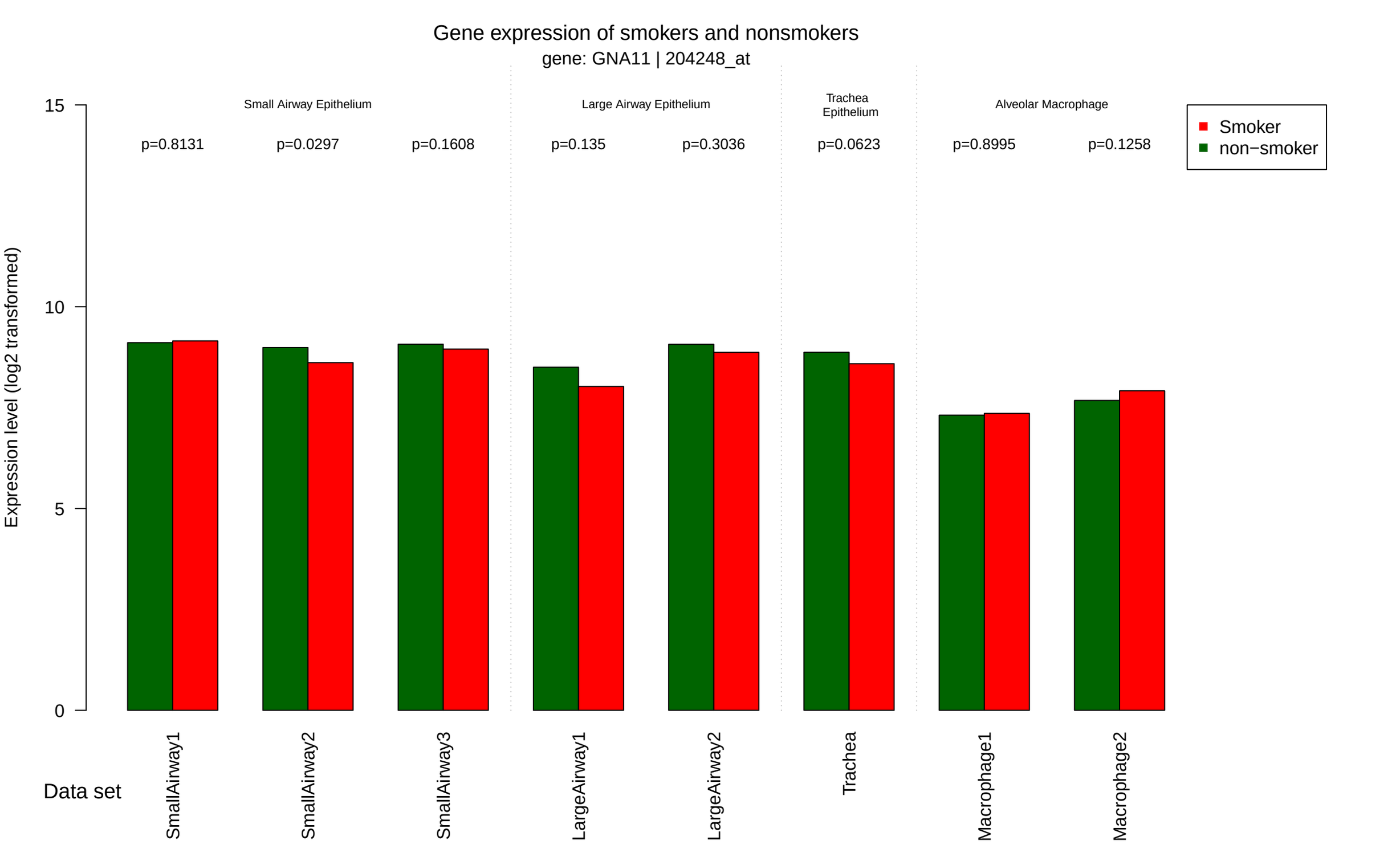
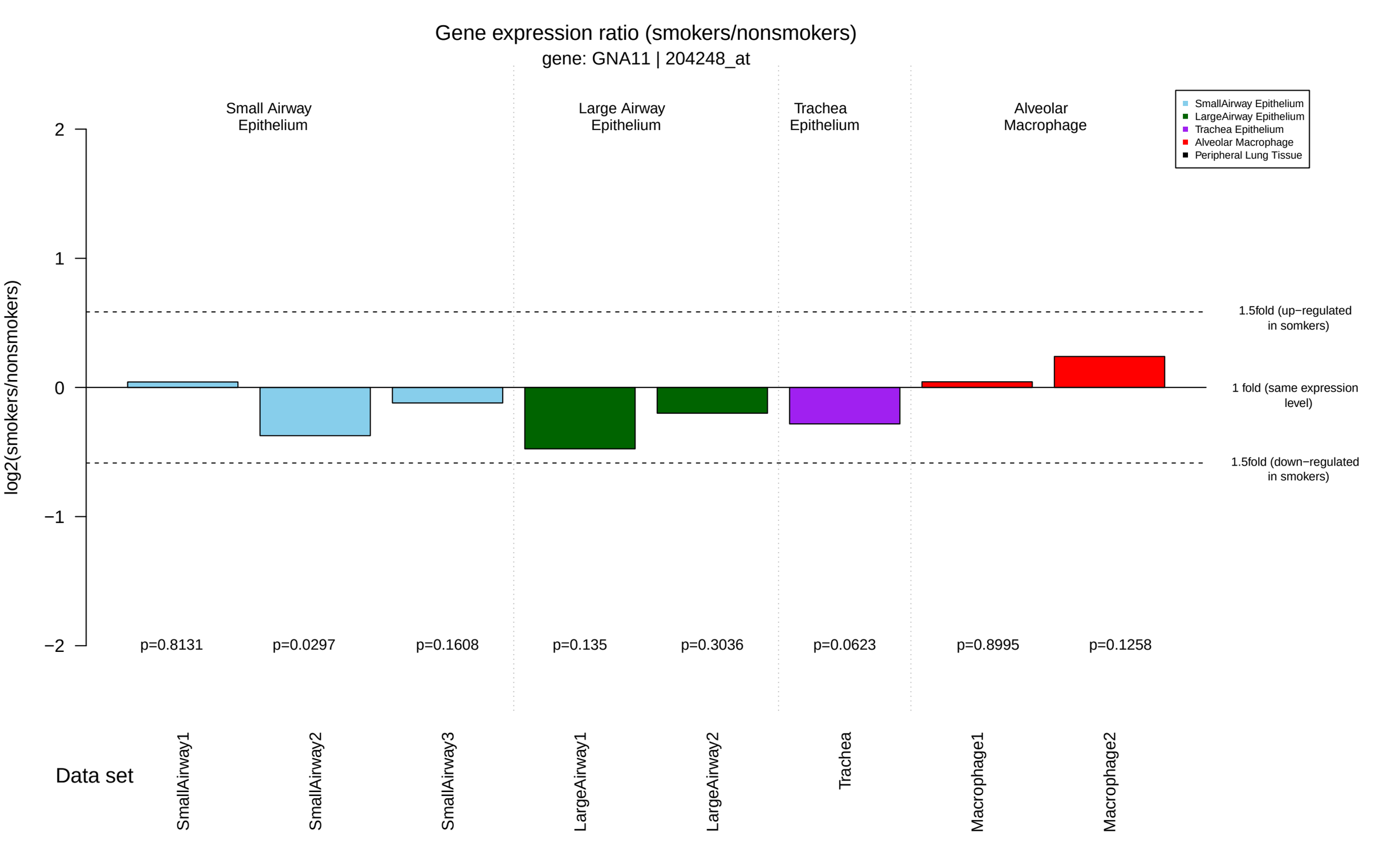
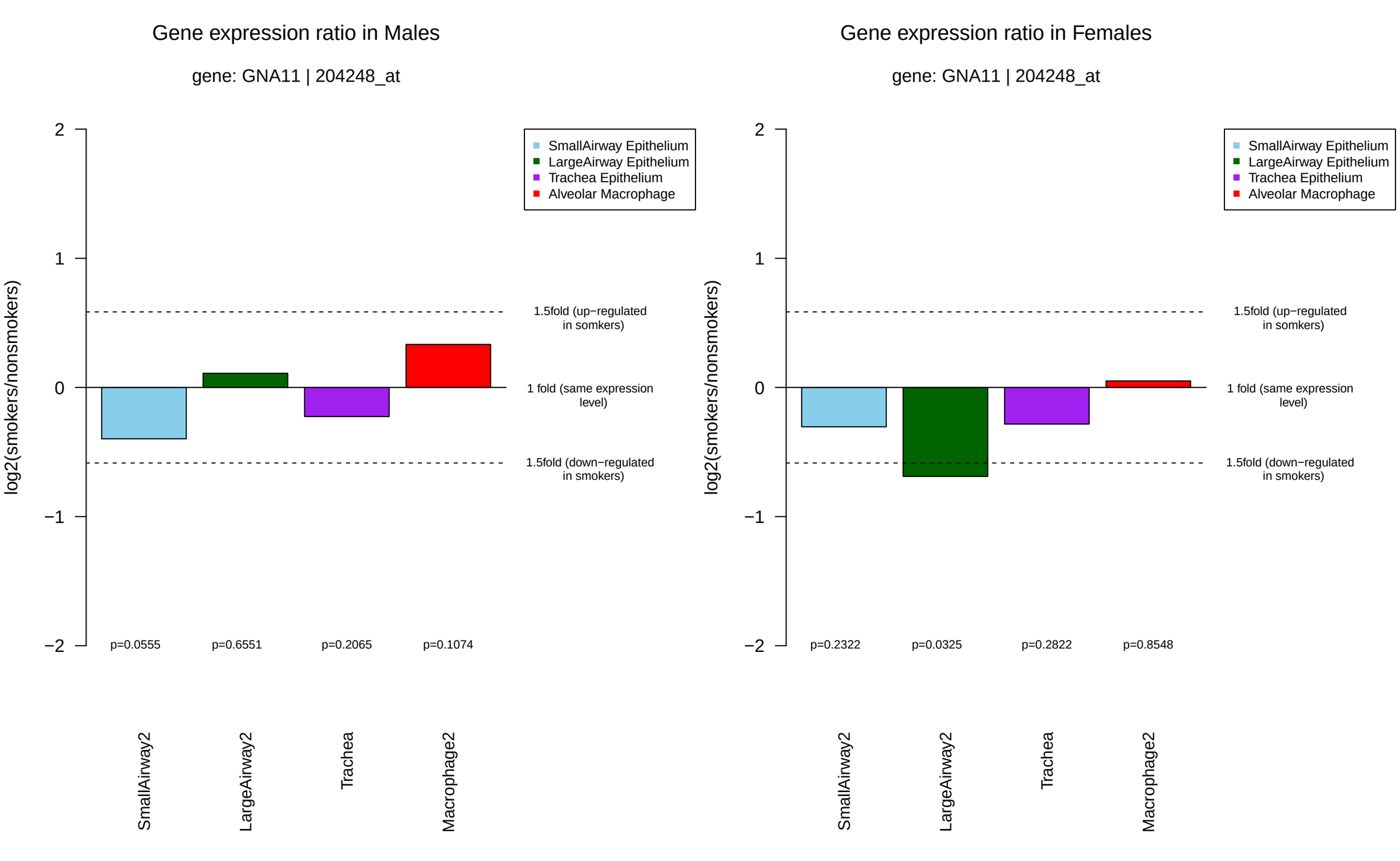
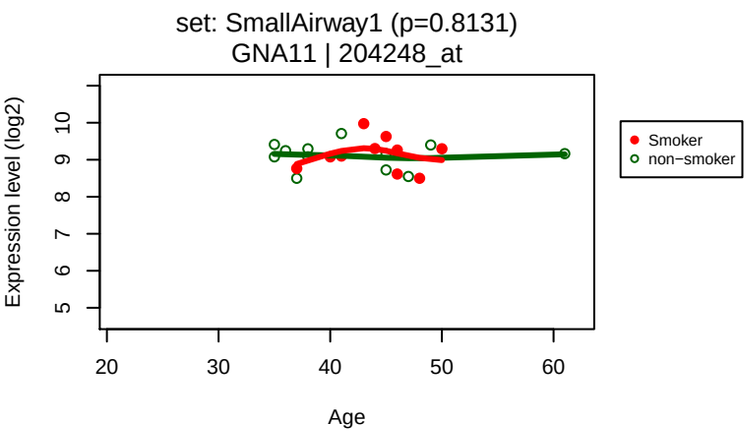
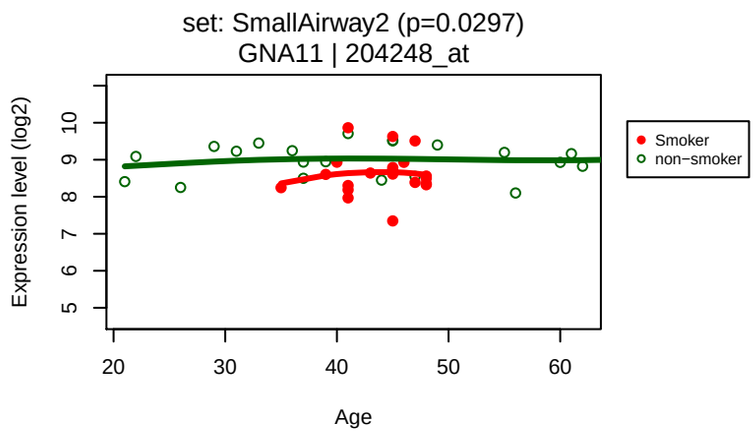
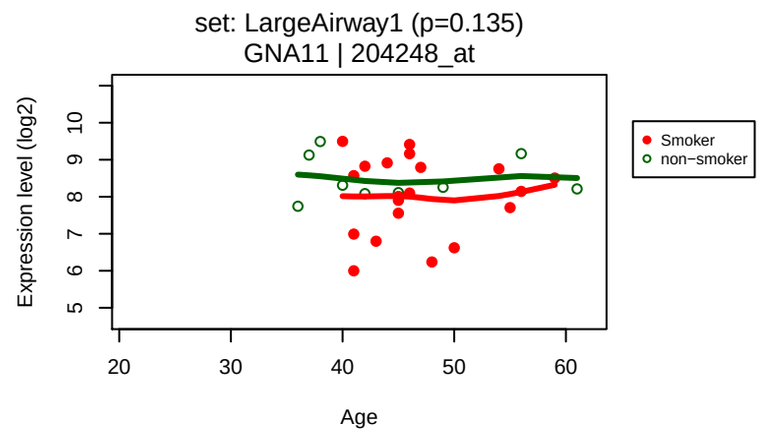

Gene expression of smokers and nonsmokers
gene: GNA11 | 204248_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8131
p=0.0297
p=0.1608
p=0.135
p=0.3036
p=0.0623
p=0.8995
p=0.1258
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

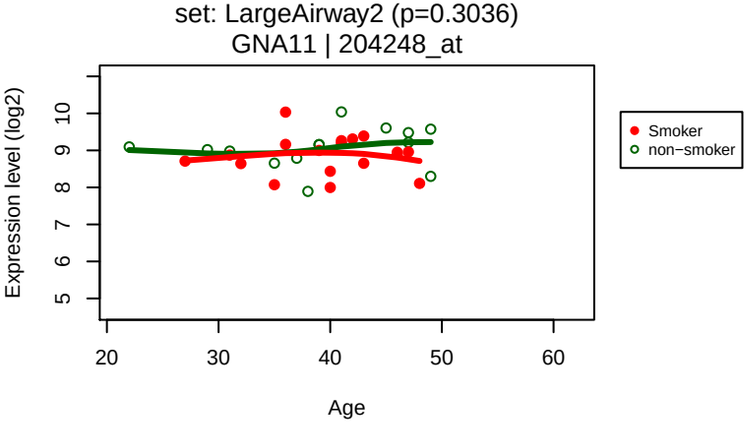

Gene expression ratio (smokers/nonsmokers)
gene: GNA11 | 204248_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8131
p=0.0297
p=0.1608
p=0.135
p=0.3036
p=0.0623
p=0.8995
p=0.1258
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

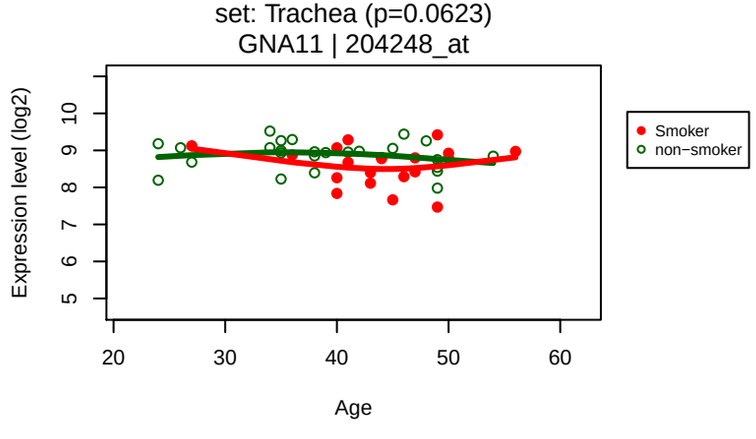

Gene expression ratio in Males
gene: GNA11 | 204248_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0555
p=0.6551
p=0.2065
p=0.1074
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNA11 | 204248_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2322
p=0.0325
p=0.2822
p=0.8548
SmallAirway2
LargeAirway2
Trachea
Macrophage2

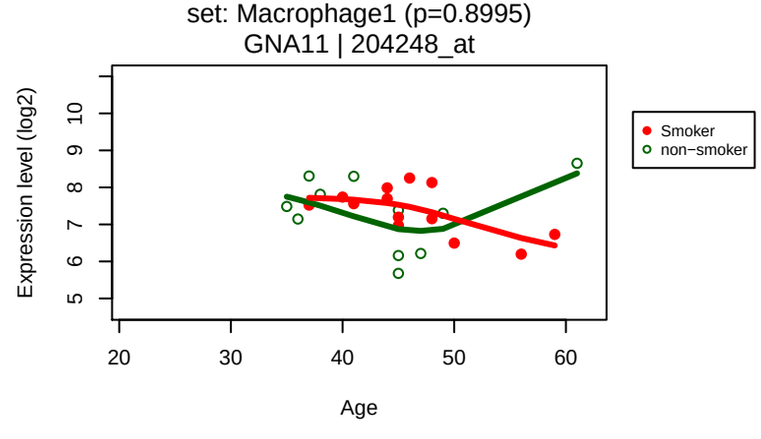

set: SmallAirway1 (p=0.8131)
GNA11 | 204248_at
Expression level (log2)
Age
Smoker
non−smoker

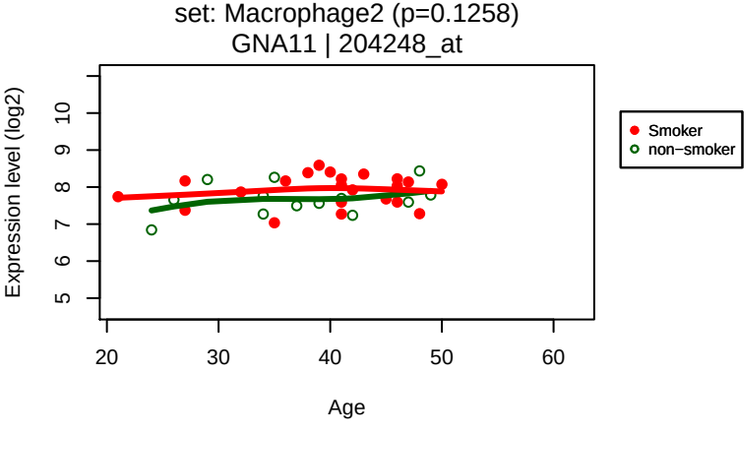

set: SmallAirway2 (p=0.0297)
GNA11 | 204248_at
Expression level (log2)
Age
Smoker
non−smoker

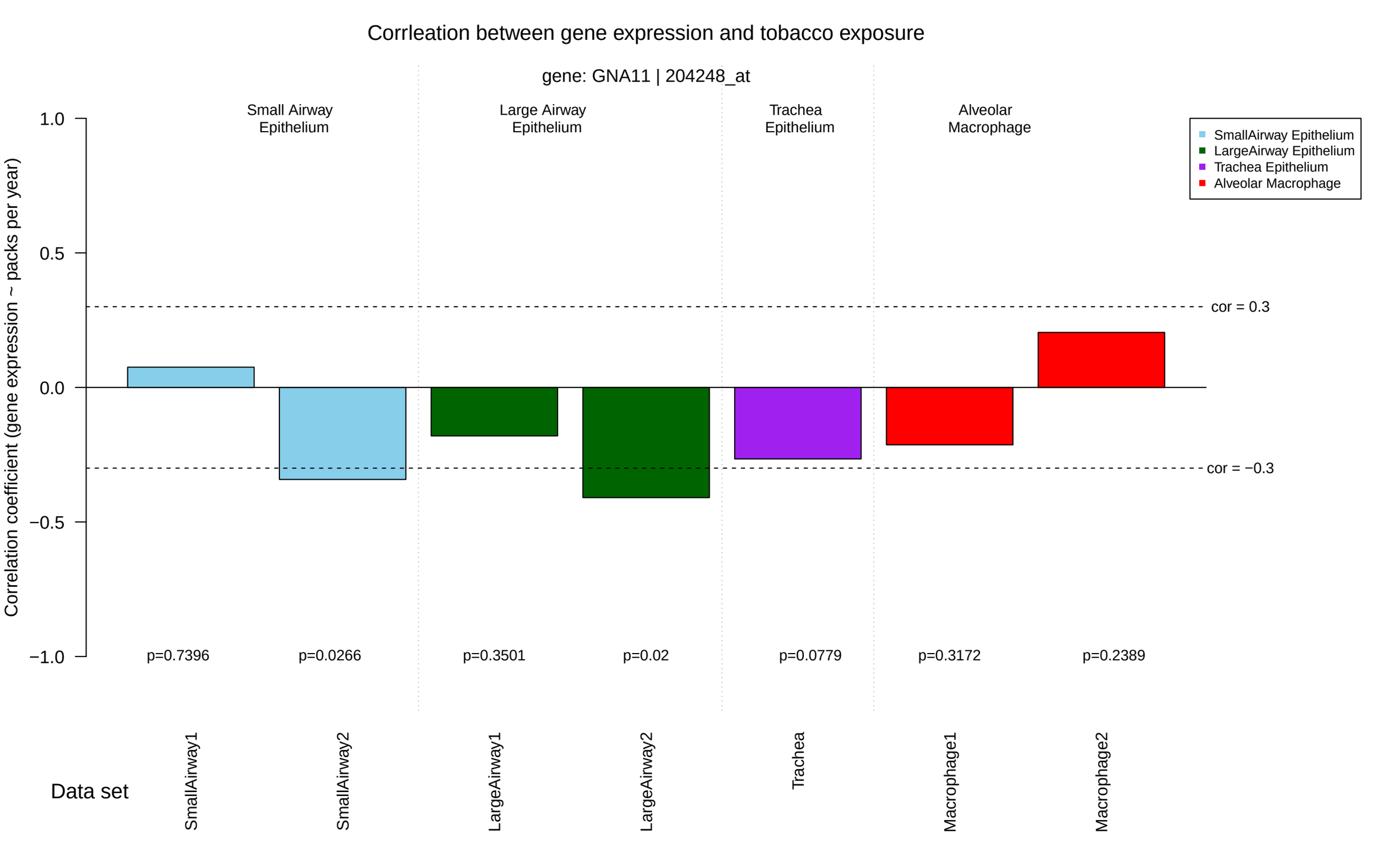

set: LargeAirway1 (p=0.135)
GNA11 | 204248_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3036)
GNA11 | 204248_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0623)
GNA11 | 204248_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8995)
GNA11 | 204248_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1258)
GNA11 | 204248_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNA11 | 204248_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7396
p=0.0266
p=0.3501
p=0.02
p=0.0779
p=0.3172
p=0.2389
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2